

UNIT 5 Health and Exercise (I)

Health and Exercise (I)

Unit Goals

- Use proper tenses in academic English
- Replace multi-word verbs with single-word verbs
- Conduct questionnaires/surveys
- Understand procedures and process
- Write a methods section

- Lesson 1

- Lesson 2



Lesson 1

- **Lead-in**
- **Reading**
- **Comprehension Check**
- **Micro-skills**
- **Macro-skills**



Lesson 1 Lead-in

Creating a survey

A survey is a research method used for collecting data from a group of respondents to gain information about various topics of interest. The following is an example of survey on food preferences.



Lesson 1 Lead-in

A survey

1. Which type of food do you prefer to have? (choose all that apply) ()

A. Deep fried

B. Wok

C. Steamed

D. Baked

E. Hot-pot (boiled)

F. Other (please specify)

2. Which style of restaurants do you prefer? (choose all that apply) ()

A. Hot-pot

B. Chinese wok

C. Western style

D. Fast food

E. Kebab

F. Other (please specify)



Lesson 1 Lead-in

A survey

3. What do you take into consideration when you eat out? (choose all that apply)
()

- A. Health and nutrition
- B. Service
- C. Convenience
- D. Cleanness
- E. Surroundings and environment

4. Please prioritize your preference on food quality: (rate the most important factor 1, and the least factor 5)
()

- A. Taste
- B. Fresh or not
- C. Appearance
- D. Spices
- E. Quantity



Lesson 1 Lead-in

A survey

5. What is your major concern in terms of GM food? (choose all that apply) ()

A. Health

B. Taste

C. Price

D. Food variety

E. Food shortage

F. Other (please specify)

Now collect answers in the class, analyze the statistics and report the result to the class.



Lesson 1

Reading

Text A

Psychological Determinants of Willingness to Taste and Purchase Genetically Modified Food (Extract 1)

2. Methods

2.1. Participants

Para. 1 A total of 100 participants (54 females and 46 males), aged 18–86 years (SD 12.6) were recruited via posters and newspaper ads from a city in the East Midlands region of the United Kingdom to take part in a psychology experiment that would involve “tasting some apples and filling in some questionnaires.” Hence the enrolment was “topic-blind,” in that the participants were unaware that the focus was on GM food, thus preventing biased recruitment. On average, the experiment took 20 minutes to complete and participants received £4 for taking part.



Lesson 1

Reading

2.2. Procedure and Materials

Para. 2 Three Red Delicious apples labeled A, B, and C served as the stimuli for this experiment. Participants read the opening paragraph of a study booklet explaining that they would be required to taste three apples and decide which of them was (1) traditionally grown, (2) organically grown, or (3) GM. In fact, all the apples had been traditionally grown. (This deception was necessary for the design of the study and participants were fully debriefed on completion of the experiment.) Participants were told that they did not have to include the GM apple in the taste test if they did not want to, and that not tasting it was of no consequence to the study, and would not jeopardize it. We reassured participants that there was no pressure to taste the “GM apple” and reinforced the point that the study was concerned with taste discrimination, and so it would not matter if the “GM apple” was not included in the test. We said we realized that some people might not want to eat GM food and the test was set up so that this would not matter. If a participant did decide not to taste the “GM apple,” one of the apples on display was removed, at random, from the taste test set-up by the experimenter. With the “GM apple” removed from the test, tasting of the other two apples could proceed.



Lesson 1

Reading

Para. 3 Two versions of the booklet had been created: in one the GM apple was said to have been modified to help prevent dental decay (health benefit), in the other the modification was supposed to have the effect of delaying the onset of rotting in the apple (money-saving benefit). Participants were randomized to versions of the booklet.

Para. 4 After signing a consent form that made it clear that they were free to withdraw from the study at any point, participants were instructed by the booklet to complete the stress-arousal check list (SACL) (Mackay et al., 1978), which measured both state negative incidental affect (stress) and arousal. The SACL, which provides a highly reliable assessment of self-reported mood, asks for responses to 30 mood-describing adjectives derived from a number of factor analytic studies (Gotts & Cox, 1988). Participants indicate to what degree each adjective describes their feelings at that moment in time (definitely, more or less, cannot decide, does not describe). Two self-reported measures of mood are provided. Stress, which relates to feelings of pleasantness- unpleasantness, is measured with such adjectives as nervous, tense, bothered, and uncomfortable. Arousal is related to physiological and behavioral states of attentiveness and wakefulness, and is measured with adjectives like alert, awake, lively, and attentive.



Lesson 1

Reading

Para. 5 The SACL was chosen to measure state mood because it ensures minimum participant intrusion, being short and easy to administer, with instructions that are easy to understand. Also, the validity of the SACL has been thoroughly tested and has been proven on its ability to differentiate between groups differing in levels of stress and arousal (King et al., 1983). The scale has excellent reliability, with internal consistencies ranging from 0.77 to 0.90 for stress and 0.74 and 0.90 for arousal (split-half coefficients). Another reason for choosing the SACL was that it has been utilized in other studies of risk and stress, for example, Kerr and Vlaminkx (1997) used it to examine gender differences in reactions to a stressful situation.

Para. 6 Participants then rated each apple for appearance, then tasted them and rated the taste as well. (All ratings were done on 7-point Likert-type scales.) They then had to decide which of the apples was traditional, organic, or GM. After the taste test, non-GM tasters were asked in the response booklet why they chose not to taste the “GM apple.” They then completed the SACL for a second time.



Lesson 1

Reading

Para. 7 The next step was to measure the attitudes associated with GM. Participants rated “GM plants” and “GM animals” on 26 7-point semantic differential scales (with a score of 7 indicating high risk, high dread, etc., 1 indicating low risk, low dread, etc., and a score of 4 indicating a neutral position). The use of such scales has become standard in risk perception research (e.g. Slovic, 1987). About half of the scales were “target items” and the others were “dummy items.” Seven of the target scales used have been shown in past research to be important in explaining perceived risk from technologies. These were: (1) dread — not dread, (2) risk increasing — risk decreasing, (3) high — low risk to future generations, (4) risky — safe, (5) globally catastrophic — not globally catastrophic, (6) vivid topic — dull topic, and (7) affects me — does not affect me. A further five target scales were specifically relevant to GM food: (1) unnatural — natural, (2) unethical — ethical, (3) high — low risk to future generations, (4) not scientifically understood — scientifically understood, and (5) disgusting — not disgusting. Two of these measures also served as indicators of integral affect (feelings toward GM food), namely, the dread and disgusting scales. Fourteen “dummy” scales were also rated, meaning that participants were unaware that the focus was on the target measures listed above.



Lesson 1

Reading

Para. 8 Participants then completed the Intolerance of Uncertainty Scale (IUS), and then provided demographic data. The IUS was used to measure trait negative incidental affect, that is, worry, which is thought to be closely related to stress (Mathews, 1990). It has excellent internal consistency and good test-retest reliability (Buhr & Dugas, 2002). In addition, the IUS seems especially relevant in the domain of GM food because one of the main concerns reported in past studies about applications of biotechnology in food is uncertainty about its effects on health and the environment, both now and in the future. Thus, it is thought that those who are predisposed to be intolerant of uncertainty, as measured by items such as “uncertainty keeps me from sleeping soundly,” “the ambiguities in life stress me,” and “uncertainty makes life intolerable,” will feel that GM technology is particularly risky.

Para. 9 Finally, after all other ratings had been completed, participants were asked whether or not they would buy GM food if it became available in shops. This was a dichotomous (yes-no) question.

(1120words)



Lesson 1

Reading

References

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Extracted from *Risk Analysis*, Vol. 24, No. 5, 2004



Lesson 1 Comprehension Check

I. Reading for main ideas

Flow chart represents the sequence of actions in a particular process or activity. Skim the text and fill in the blanks to complete the flow chart.

Three apples were labeled A, B, and C.

1. Booklets had been created.

2. Checklist was completed.

Each apple was rated for 3. appearance and the taste.

The 4. attitudes associated with GM were measured.

The Intolerance of Uncertainty Scale (IUS) was completed, and then 5. demographic data was provided.

Finally, participants were asked if they were willing to buy GM food if they could choose.



Lesson 1 Comprehension Check

II. Reading for evidence

Do the following statements agree with the claims of the writer in Text B?

YES *if the statement agrees with the claims of the writer*

NO *if the statement contradicts the claims of the writer*

NOT GIVEN *if it is impossible to say what the writer thinks about this*

NO 1. The purpose of the experiment was to find out about taste discrimination.

YES 2. The participants could refuse to taste GM apple if they didn't like it.

YES 3. SACL evaluates extent to which some adjectives indicated the feeling of the participants.

NO 4. The SACL was used because it ensures participant intrusion.

NG 5. The participants were not told the main purpose of the experiment, so that they would give true answers.

YES 6. IUS is especially useful in GM food, because people are not sure about its effect on health and environment.



Lesson 1 Micro-skills

I. Tenses in academic English

II. Single-word verbs and multi-word verbs



Lesson 1 Tenses in academic English

Simple and perfect, present and past tenses are used most in academic writing.

Present simple

- A general principle
- Someone's opinion
- The results of an experiment
- A habitual action in the present
- A theory
- A claim
- A fact

Past simple

- What happened
- What happened in the past but is different now
- Ideas that were held in the past but are no longer held

Present perfect

- Something that has changed over time
- Recent research and its present impact



Lesson 1 Tenses in academic English

- A. Underline the present simple, past simple and present perfect in paragraph 4, and explain the function of the tenses.**



Lesson 1 Tenses in academic English

B. Can you explain why each of the following 10 sentences was written in a particular tense? Choose your reasons from the box:

Example:

Smith and Jones think this is unlikely.

Answer:

present simple – someone's opinion



Lesson 1 Tenses in academic English

a general principle	the results of an experiment
what happened in an experiment	a habitual action in the present
ideas that were held in the past	recent research and its present impact
something that has changed over time	a theory
a fact	a claim

1. The charcoal was ground and put in a container.
2. These findings indicate the danger of mixing potassium nitrate, charcoal and sulphur.
3. General relativity is a theory of great significance.
4. Advertisers work on the principle that most people like chocolate.
5. Recent surveys have shown that most people prefer milk chocolate to white chocolate.
6. Many people eat chocolate every day, as it is a good source of antioxidants.
7. Many people have gained weight quickly because of eating increasing amounts of chocolate.
8. It was thought that chocolate would replace alcohol as a beverage.
9. Chocolate is made from cocoa beans.
10. Smith maintains that white chocolate is not really chocolate.



Lesson 1 Tenses in academic English

Answers:

1. Past simple – what happened in an experiment
2. Present simple – the results of an experiment
3. Present simple – a theory
4. Present simple – a general principle
5. Present perfect – recent research and its present impact
6. Present simple – a habitual action in the present
7. Present perfect – something that has changed over time
8. Past simple (passive) – an idea that was held in the past
9. Present simple (passive) – a fact
10. Present simple – a claim



Lesson 1 Tenses in academic English

C. There are five mistakes in the following paragraph, one in each numbered line. Find the mistakes and write the correct answers in the blanks.

Women on average live longer than men, which seems to suggest that women also age slower than men. However, the potential difference in the pace of aging between the sexes are relatively controversial topic, and both positions, i.e. “men age faster” and “men and women age at the same pace”, found some support. We therefore employed parametric models previously established in model organisms as well as two nonparametric approaches to compare the pace of aging between the sexes using freely available mortality data from 13 high-income countries. Our results supported the hypothesis that men age faster than women while also suggesting that the difference is small and that from a practical standpoint male mortality rates are behaving similarly to the mortality rates of women approximately eight years their senior.

1. ~~are~~ → **is**
2. ~~found~~ → **have**
3. ~~found~~ —
employed →
employ
4. **supported** →
support
5. **are behaving**
→ **behave**



Lesson 1 Single-word verbs and multi-word verbs

English often has more than one choices to express things. The choice is often between a **multi-word verb** or **phrasal verb** and a **single-word verb**. In everyday spoken English, a phrasal verb is often preferred, while for written academic English, a single verb is used when possible.



Lesson 1 Single-word verbs and multi-word verbs

A. Match the single-word verbs in the box to the multi-word verbs (1-10)

arise check conduct consider constitute
delay discover discuss highlight investigate

1. come up → arise
2. look into → investigate
3. put off → delay
4. make up → constitute
5. carry out → conduct
6. go into → discuss
7. point up → highlight
8. go through → check
9. think about → consider
10. find out → discover



Lesson 1 Single-word verbs and multi-word verbs

B. Underline the multi-word verbs in the following sentences and replace them with single-word verbs.

1. In this book Sergeant goes into the causes of the Civil War in some depth. **discusses**
2. The psychologist pointed out that it was very unusual for a young child to behave in this way. **observed**
3. Please go back over this term's notes. **revise / review**
4. In her latest article Kaufmann puts forward a theory which is likely to prove controversial. **presents**
5. We worked out that software alone accounted for about one-third of the overall cost of the project. **calculated**



Lesson 1

Macro-skills

I. Writing a methods section

The **methods section** provides the information by which the validity of a study is ultimately judged. Therefore, the author must provide a clear and precise description of how an experiment was done, and the rationale for the specific experimental procedures chosen. It must be written with enough information so that: (1) the experiment could be repeated by others to evaluate whether the results are reproducible, and (2) the audience can judge whether the results and conclusions are valid.

The methods section is usually composed of three subsections, i. e. participants, measures and procedures. The methods section may be brief when the methods used are familiar and standardized, or quite lengthy, when the methods used are new and complex and thus require careful elaboration. Therefore, there are two models of the methods section: one without subheadings when it is short and relatively simple and the other with subheadings when it is long and complex.



Lesson 1

Creating an outline

A. Read the following methods section carefully, and answer the questions listed below.

The paper presents complex interdisciplinary analysis of social, psychological, legal and economic aspects of intellectual property creating and protecting. The primary method of research is the panel survey of creative persons, researchers and creators, professors and lecturers. By means of panel survey different forms of motivation for intellectual property creating and protection were revealed. Special attention is paid to creative work of lecturers at universities and the attitude to intellectual property creating and protecting.

The aim of the panel survey was the identification of the most important factors for development of creative and / or scientific research activities and the main problems of motivation of intellectual property infringement and protection.

Through the panel survey, questionnaires were spread among 100 researchers, professors and lecturers of Perm State National Researches University and Perm National Researches Polytechnic University (Russia). The questionnaire of panel survey was conducted from February to June 2014.



Lesson 1

Macro-skills

Questions:

1. How many people participated in the survey and what did they do by profession?
2. What was the purpose of the survey?
3. How did the researchers conduct the survey?
4. What was revealed by the survey?

1. Questionnaires were spread among 100 researchers, professors and lecturers of Perm State National Researches University and Perm National Researches Polytechnic University (Russia).
2. The identification of the most important factors for development of creative and / or scientific research activities and the main problems of motivation of intellectual property infringement and protection.
3. The primary method of research is the panel survey of creative persons, researchers and creators, professors and lecturers.
4. Different forms of motivation for intellectual property creating and protection were revealed



Lesson 1

Creating an outline

B. Read the first two paragraphs of Text A again, and compare them with the following notes. Then fill in the blanks with appropriate expressions from the paragraphs.

In Text A, the methods section is composed of participants, procedure and materials.	
detailed information	Participants 1 <u>Age</u> range: 18 - 86 - Where they are from - What they would do * unaware of the 2 <u>focus</u> — avoid biased recruitment
clear enough to be repeated? results reproducible?	Procedure and Materials - Three labeled 3 <u>apples</u> - a study 4 <u>booklet</u> explaining the requirements - no 5 <u>pressure</u> to taste the “GM apple” – not matter to the test - the “GM apple” removed from the test if the participants want
very complex section, needs careful and detailed description to be convincing	



Lesson 1

Macro-skills

II. Surveys

The **survey** method is a technique of gathering data by asking questions to people who are thought to have the desired information. Survey methods include telephone surveys, face-to-face interviews, surveys by mail, and Internet questionnaires.



Lesson 1

Surveys

C. Complete the following passage using the proper word from the box.

cheapest	considerations	questionnaires	responses	several
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Each survey method has its advantages and disadvantages, which should be weighed against the budget, available personnel and other 1 considerations.

In general, face-to-face interviews, although they are costly and time-consuming, deliver the most representative results and finest granularity in 2 responses. On the other hand, online 3 questionnaires sometimes come with significant biases, but they are the easiest and 4 cheapest surveys to administer.

If you're only going to rely on one form of survey, such as an online questionnaire, consider surveying more people to make up for possible biases. For the cleanest results, you might want to administer 5 several different kinds of surveys.



Lesson 1

Surveys

D. Match the definition of some survey questions and the examples.

Here are definitions of some most common survey questions:

1. Structured questions ask a question and then provide answer choices below.
2. Unstructured questions remove predetermined answers from the equation.
3. A “rating” question seeks to answer how respondents would rate their experience on a scale.
4. A “ranking” question gets an ordered list of preferences.



Lesson 1

Surveys

D. Match the definition of some survey questions and the examples.

Here are examples of the above survey questions. Match the type of question with the examples.

a. “The Brooklyn Zoo is fun for both children and adults.” 3

(a) Strongly Disagree

(b) Disagree

(c) Agree

(d) Strongly Agree

b. “Talk about your first experience of stepping into an Apple Store.” 2



Lesson 1

Surveys

D. Match the definition of some survey questions and the examples.

- c. “In the space provided below, rank the brands you trust the most, with ‘1’ being the most trustworthy and ‘5’ being the least trustworthy.” 4
- (a) ___ McDonald’s
 - (b) ___ Google
 - (c) ___ Walmart
 - (d) ___ Costco
 - (e) ___ Apple
- d. “What is your favorite online activity?” 1
- (a) Chatting / IM
 - (b) Social Networking
 - (c) Knowledge Sharing / Forums
 - (d) Shopping / E-commerce



Lesson 2

- **Lead-in**
- **Reading**
- **Critical Thinking**
- **Micro-skills**
- **Macro-skills**



Lesson 2

Lead-in

Events of the Physical Activity Courses

A. Look at the sports categories and examples. Try to add more examples to each category.

aquatics	dance & rhythms	fitness & conditioning	outdoor activities	sports
sailing swimming	shag swing	self-defense weight training	equestrian rock-climbing	basketball fencing

B. Choose from the above table one sport course that you have enrolled in, and introduce it to the class. Explain why you take the course.



Lesson 2

Reading

Text B

Activity Levels of College Students Enrolled in Physical Activity Courses (Extract 1)

Methods

Participants and setting

Para. 1 Participants in this study were college students enrolled in PA courses, which are part of a campus wide PA program at a large university located in a mid-sized city in the Southeastern United States. The program, which offers more than 50 different courses with 130 different course sections taught by faculty, graduate students, and adjunct professors, serves 4,200-4,500 students a year. The primary goal of the program is to help students become physically educated so that they may pursue a lifetime of health and PA. The courses offered fit into five different categories, including aquatics (e.g. sailing, swimming), dance and rhythms (e.g. shag, swing), fitness and conditioning (e.g. self-defense, weight training), outdoor activities (e.g. equestrian, rock climbing), and sports (e.g. basketball, fencing). There is no university-wide policy that requires students to take a PA course to meet graduation requirements and only two degree programs (i.e. physical education and dance) that require students to take certain courses in the PA program.



Lesson 2

Reading

Para. 2 During the fall of 2016 academic semester, there were 20 different courses with 76 different course sections which met this study's criteria (see below). There were 1,199 students enrolled in these courses. From these courses, a purposive sample of nine classes was selected. Classes selected (a) were held on campus in the main facility used for PA courses (due to travel, time and cost constraints of this unfunded study), (b) had a minimum of eight participants enrolled (PA classes of fewer than eight students at the start of the semester were at greater risk of being dropped for low enrollment), (c) focused on land-based activities (due to the researchers not having access to waterproof accelerometers), (d) met for the full academic semester (allowing for 6 evenly dispersed data collection intervals), and (e) were scheduled at different times from one another (due to a limited number of researchers to collect data). Each class selected focused on different content, including swing dance, spinning, flying disc sports (a combination of disc sports, such as Ultimate Frisbee and Frisbee golf), archery, Tai Chi, weight training, yoga, karate, and jogging. Classes met at least twice per week for 50 or 90 minutes per class period.



Lesson 2

Reading

Para. 3 Of the 130 total students enrolled across the nine classes, there were 66 students who participated (M—age=20.58, SD=2.04, 38 female). These students self-identified their race/ethnicity as 74% Caucasian, 10% Asian/Pacific Islander, 7.5% African-American, 4.5% Hispanic/Latino, and 3 % Other. Participants ranged from their first year of college through graduate school and represented nine of the 13 schools and colleges within the university. Mean body mass index (BMI) was 24.85 ± 4.08 kg/m² for male participants and 23.59 ± 4.59 kg/m² for female participants. A total of 39% (n=11) of male participants and 32% (n=12) of female participants were classified as either overweight or obese. Thirty-six students reported having participated in another college PA course in the same program and seven reported having participated in another college PA course at a different college or university. All of the participants had previous participation in sporting activities. For demographic information, see Table 1 and Table 2.

Lesson 2

Reading

Table 1 Participant Demographics

	Males [⌘] (n=28) [⌘]	Females [⌘] (n=38) [⌘]	Total [⌘] (n=66) [⌘]
Age (SD) [⌘]	20.79(2.86) [⌘]	20.66 (1.95) [⌘]	20.58 (2.04) [⌘]
BMI kg/m ² (SD) [⌘]	24.85 (4.08) [⌘]	23.59 (4.59) [⌘]	23.54(6.375) [⌘]
Under weight (%) [⌘]	0 (0.00) [⌘]	2 (5.26) [⌘]	2 (5.26) [⌘]
Normal weight (%) [⌘]	17 (60.71) [⌘]	24 (63.16) [⌘]	41 (92.24) [⌘]
Over weight (%) [⌘]	8 (28.57) [⌘]	8 (21.05) [⌘]	16 (24.81) [⌘]
Obese (%) [⌘]	3 (10.71) [⌘]	4 (10.53) [⌘]	7 (10.62) [⌘]

Lesson 2

Reading

Table 2 Class and Participant Characteristics

Class	Spinni	Flyin	Archer	Tai	Weight	Yoga	Karat	Joggi	Total	
Swing	ng	g Discy	y	Chi	Trainin	e	ng			
Total	8	6	5	12	7	7	9	5	7	66
Participants										
Ethnicity										
Hispanic	0	1	0	0	0	1	1	0	0	3
Caucasian	5	4	5	8	6	6	7	3	5	49
Black	1	1	0	0	2	0	0	1	1	6
Asian	1	1	0	1	1	0	1	1	1	7
Other	1	0	0	1	0	0	0	0	0	2



Lesson 2

Reading

Table 2 Class and Participant Characteristics

Why Taking Course											
Requirement	0	1	2	5	1	3	0	0	1	13	
Knowledge	7	2	0	6	3	1	3	2	2	26	
Increase PA	1	5	2	1	0	4	5	2	5	25	
Weight Control	0	1	0	0	0	0	0	1	2	4	
Other	2	0	1	2	3	0	1	2	0	11	
Year in College											
One	0	1	2	0	1	1	0	0	0	5	
Two	3	1	1	2	1	2	4	0	3	17	
Three	1	0	1	3	2	3	2	2	1	15	
Four	3	2	1	3	3	0	3	3	3	21	
Five	1	1	0	3	0	0	0	0	1	5	
Graduate	0	1	0	1	0	1	0	0	0	3	



Lesson 2

Reading

Procedures and instrumentation

Para. 4 The University Institutional Review Board, the director of the PA program, and the instructors of the selected classes granted permission for the researchers to conduct the study. Students in these classes were asked to volunteer to be a part of the study, and volunteers consented to being part of the study. Participants were enrolled in the study until a minimum of 50% of the class volunteered. We chose a minimum of 50% because of the limited number of accelerometers available for use in the study, and because of the varying class sizes in this study. Students did not receive any incentive for their participation. Participation included completing a questionnaire on the first day of the study and wearing an accelerometer affixed to the right hip on 6 preselected dates during the semester. A trained data collector, delivered accelerometers to classes on the days accelerometers were to be worn, and was responsible for ensuring proper wear location during observation periods and recording “time on” and “time off”.



Lesson 2

Reading

Para. 5 Trained data collectors collected data at six time points, including the weeks of August 29, September 12, September 26, October 17, October 31, and November 14. In all but three cases, data for each time point for each class were collected during the same week. Data was collected during the following week for all three exceptions. On the first day of data collection, participants filled out a 13-question background questionnaire, which included demographic questions (e.g. age, height, race, weight,) and a single question that focused on participants' reasons for en-rolling in the PA class ("Why are you taking this course?"). Five response options were provided: (a) "fulfill a requirement," (b) "gain knowledge of the sport/activity," (c) "increase physical activity," (d) "weight control," and (e) "other." Participants were prompted to choose all responses that applied to them. On each data collection day, participants wore an ActiGraph GT3X+ accelerometer (Shalimar, Florida) on their right hip for the duration of each PA class. Time on and time off was noted on a data collection sheet to monitor participants' accelerometer wear time..



Lesson 2

Reading

Data Analysis

Para. 6 Descriptive and demographic data were presented as means and standard deviations. Accelerometers were downloaded at the completion of each data collection week using ActiLife (version 6.13.3) software. Validated adult hip-based cut points (Freedson, Melanson, Sirad, 1998) of accelerometer counts per 10 seconds were used to distill sedentary (<16.5), light (16.6-325.1), moderate (325.2-954), and vigorous intensity (>954.1) PA, respectively. All analyses of PA data were performed using STATA (v. 13.0, College Station TX, [Freedson et al., 1998]). Participants total accumulated PA was summed and averaged for each class period observed. Frequencies were calculated for participants' responses to the single questionnaire item pertaining to reasons for enrolling in PA courses.

(1298 words)

(Extracted from *American Journal of Health Studies* 32(3) 2017)



Lesson 2 Critical Thinking

Read the text again and answer the following questions.

1. Do you agree that university requirement is a major motivation for students to participate in sports courses? Give your reasons.
2. The procedure part of a research paper could be complex. What do you think of the procedure part of this extract? Explain.
3. Compare the sports courses mentioned in the study, and those in your university. What are the differences? What causes the differences?



Lesson 2

Micro-skills

I. Acronyms

II. Flow Chart

III. Collocation (1)



Lesson 2

Acronyms

Acronyms are often used in academic writing in order to avoid the repetitive use of long, cumbersome titles. Acronyms are defined as words formed by the first letters of words in a name or title.

For example, in Text A, *Genetically Modified Food – GM food*, the *stress-arousal check list (SACL)*, and in Text B, *Physical Activity Courses – PA courses*.



Lesson 2

Acronyms

A. Here are some commonly used acronyms of international organizations. Write the full names.

1. UNESCO



United Nations Educational, Scientific And Cultural Organization
(联合国教科文组织)

2. IMF



2. International Monetary Fund (国际货币基金组织)

3. OPEC



3. Organization of Petroleum Exporting Countries (石油输出国组织)

4. IOC



4. International Olympic Committee (国际奥林匹克委员会)

5. NASA



5. National Aeronautics And Space Administration ((美国)国家宇航局)



Lesson 2

Acronyms

B. Here are some acronyms that are not so common. Look them up and find the full names.

1. AIR American Institutes for Research
2. ASL American Sign Language
3. EL English Learner
4. SSCI Social Sciences Citation Index
5. GMT Greenwich Mean Time



Lesson 2

Flow Charts

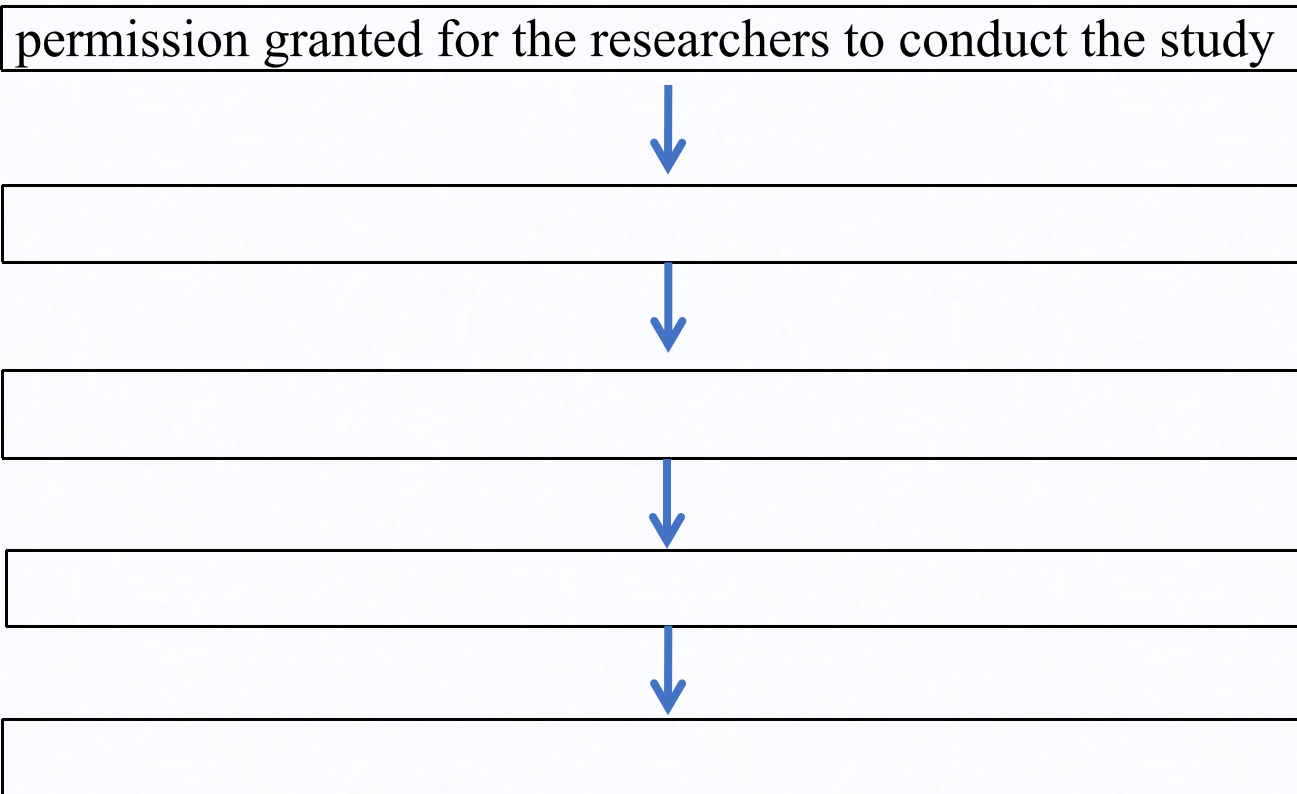
Flow chart is very important both in reading and in writing. It is an effective way of making reading notes, and it is also a writing technique when you try to describe a series of actions. Flow chart shows the connections between the different stages of a process. Before you make a flow chart of a process, you need to identify the most important information in each stage of process. To make a flow chart coherent and cohesive, you need signal words like *first*, *then*, *next*, *after this*, *finally*, etc. to connect the stages. You also use passive voice a lot because usually you don't need to mention who or what is doing an action.



Lesson 2

Flow Charts

- A. Read Paragraph 4 again, and then make a flow chart of the process. The first is already done as an example.**



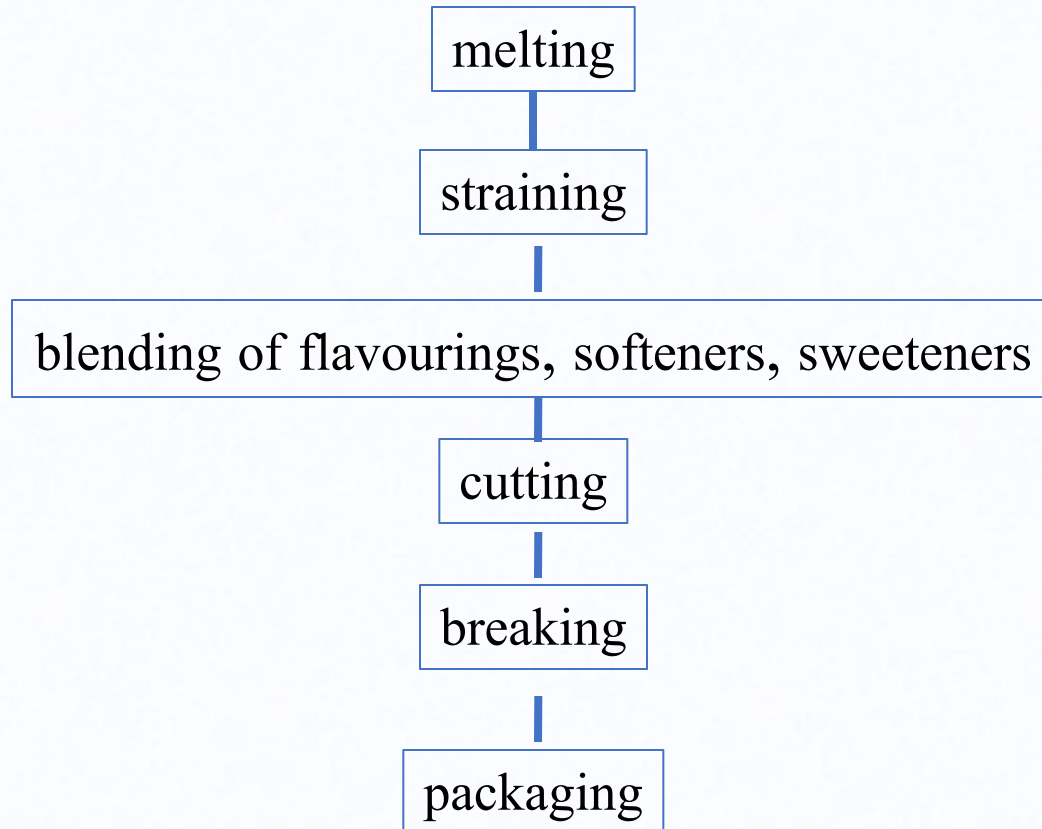


Lesson 2

Flow Charts

B. Look carefully at the following flow chart of how *chewing gum is made*.

How chewing gum is made





Lesson 2

Flow Charts

Now complete the description with the correct form of the words in the box. You may use some of the words more than once.

add	container	first	finally	heat	ingredients	liquid	machine
mixture	next	pass	pieces	place	shapes	show	then

The diagram 1 shows how chewing gum is produced. 2 First the 3 pieces of gum are put into a 4 container and 5 heated until they form a single mass. This is 6 then strained until all of the 7 liquid has been eliminated. 8 Next / Then the gum is 9 placed into another 10 container and the desired flavourings, sweeteners and softeners are 11 added. This 12 mixture is blended until all of the 13 ingredients are mixed together. The gum then 14 passes / travels to a different 15 machine which uses rollers to flatten out the gum. 16 Next / Then the thin sheets of gum are cut into rectangular 17 shapes / pieces and broken into separate 18 pieces. 19 Finally the gum 20 travels / passes along a conveyor belt where it is enclosed in packaging ready for distribution.



Lesson 2

Collocation (1)

In the field of English for Academic Purpose, **collocation** is recognized as an important feature in academic writing. In linguistics, collocation is the way that some words occur regularly whenever another word is used (Collins COBUILD Dictionary). For example, *produce* is often followed by *result*, *argument* is often modified by adjectives like *major*, *strong*, *valid*, *compelling*, etc.



Lesson 2

Collocation (I)

A. Choose proper word from the box and fill in the blanks.

further	into	doing	purposes	showed
---------	------	-------	----------	--------

1. The evidence for the causes and results of problem requires further **research**.
2. Copies of the tape can be made for **research** purposes.
3. Our **research** showed that company growth is more often determined by internal factors than by the external business environment.
4. He has carried out extensive **research** into renewable energy sources.
5. She's doing **research** on Czech music between the wars.



Lesson 2

Collocation (I)

B. There are different collocation patterns. In the table below are some collocation patterns of *research*.

<i>research</i> as a noun	adjective + research
	research + noun
	research + verb
	verb + research
	research + preposition
<i>research</i> as a verb	research + preposition
	research + noun



Lesson 2

Collocation (I)

Now observe the collocation of *research* in the above Exercise A and match the patterns with the sentences. The first is already done as an example.

1	<i>Adjective + research</i>
2	research + noun
3	research + verb
4	research + preposition
5	verb + research



Lesson 2

Collocation (I)

C. Sometimes collocation in English is different from that in Chinese. Beginners may make mistakes in this part. Now correct the mistakes in collocation usage in the following translations.

1. 开支票 open a check write out a check
2. 接电话 receive a telephone answer the telephone
3. 拥挤的交通 crowded traffic heavy traffic
4. 伤得很重 heavily injured badly injured
5. 对他很信任 have great trust for him. have great trust in him



Lesson 2

Macro-skills

Methods: Procedures and process

Procedures and process are important parts of the method section in academic writing. They should be scientific, representative and clear. Usually, they include sampling procedures, instrumentation, data collection procedures and data analysis procedures, to name only some.



Lesson 2

Method: Procedures and process

A. The following paragraph is the method part of a paper. Read it closely and then answer the questions.

Methods

Mortality rate data were acquired from www.mortality.org on 12 July 2017. The Human Mortality Database (HMD) contained data about mortality rates for 39 sovereign countries and several other smaller areas and populations. In our analysis we focused on 13 high-income, western (plus Japan), stable countries with populations exceeding 8 million. The analysed countries are: Australia, Belgium, Canada, France, Italy, Japan, the Netherlands, Portugal, Sweden, Switzerland, the United Kingdom, the United States of America and West Germany. The reasoning behind the selection criteria was to analyse only the most stable countries with a high population to minimize an effect of chance and aging unrelated effects on the outcome of our analysis. Mortality data analysed for each country were in periods starting from 1980 to 1984 to the newest available data in the Human Mortality Database. In most cases this was the year 2014. However, for the UK, Italy, and Canada it was 2013, 2012 and 2011 respectively.



Lesson 2 Drafting an introduction

1. What information is presented in this part?

Data collection procedures are presented, including which countries the data come from, and why these countries are chosen, and periods in time of these data.

2. What information is missing?

The missing parts are instrumentation and data analysis procedures.



Lesson 2 Drafting an introduction

B. Write a Methods section based on the following information provided (“I” refers to the researcher).

I sent an email to invite some medical students. Altogether there are 590 students. These students are from the Griffith University in Queensland, Australia.

All students finished a questionnaire consisting of 30 questions. The original purpose of the questionnaire is to analyze students’ understanding of the surgical subspecialties.

I gave all students 25 different surgical scenarios and asked them to choose which subspecialty will be most likely to treat surgical conditions. Altogether there 11 surgical subspecialties.

I used SPSS Statistics for Windows, Version 22.0 to analyze all the statistics. Based on Chi-squared analysis, I calculated and obtained such statistics as Univariate frequency differences, odds ratios, 95% confidence intervals, and *P*-values.



Thanks for Watching!